

Carlos Serrouya

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WORK EXPERIENCE AND PERSONAL PROJECTS

Data Science Intern *Open Ocean Robotics* 2025 – Present (1 Year)

- Key contributor in taking **Enhanced Horizon™** from development to a robust object tracking and localization system used on autonomous surface vessels.
- Deployed ML applications on edge devices and cloud using Docker, PyInstaller, and Git CI pipelines; optimized latency and throughput with TensorRT, NumPy, and multithreading
- Implemented an end-to-end model benchmarking pipeline that auto-generates reports and visualizations for each new model release, enhancing reproducibility and performance tracking
- Improved object detection models developed with PyTorch by +20% inference speed; mAP50: 28→45, mAP50-95: 7→20, precision: 40→80%, recall: 30→40%, class confidence: 80→90%
- Developed distance estimation object tracking algorithms using camera ray tracing, Kalman filter, statistical propagation of errors; deployed from simulation to edge devices on USVs with highly positive field feedback from USV operators, earning a company wide innovation award
- Automated large-scale data uploading and labeling pipelines, processing terabytes every two weeks, reducing manual upload time by 3 hours per cycle
- Worked closely with a fellow intern during Python-to-Rust migration of key DS applications, explaining statistical and mathematical concepts and assisting with debugging
- Conducted advanced and extensive statistical analysis of data science applications performance; visualized results using matplotlib, seaborn, grafana, and **Xploreview™** to guide engineering improvements and inform stakeholders. Presented as Visual lead to multiple clients totalling in hundreds of thousands in contracts.

Data Science Intern *Alberta Biodiversity Monitoring Institute* May-September 2023, 2024

- Developed a proof-of-concept application to estimate snow coverage and depth from camera trap images using a fixed-reference-pole semantic segmentation and classification approach; done by using Tensor flow for U-net model and transfer learning to random forest model for classification model. Was able to reliably estimate snow depth in a given image within 7cm
- Created object contouring, detection, and outlier detection algorithms, to identify inconsistencies in application results
- Managed large datasets with Pandas and SQL, integrated basic statistical models as SQL triggers.
- Presented machine learning results to a non-technical panel, achieving 84% snow depth accuracy and 96% coverage classification.
- Reduced false positives in Microsoft's Megadetector by 72%, false negatives by 9%, by implementing multiple regression post process model
- Visualized model performance data with Matplotlib and Seaborn.

EDUCATION AND CERTIFICATIONS

University of Calgary

BSc Computer Science 2022-2027
BSc Mathematics 2021-2026

Coursera

Advanced Learning Algorithms 2023
Regression & Classification 2023

ADVANCED SKILLS

Microsoft Office
Testing Automation

Data Validation
Data Visualization

SQL
Power BI